

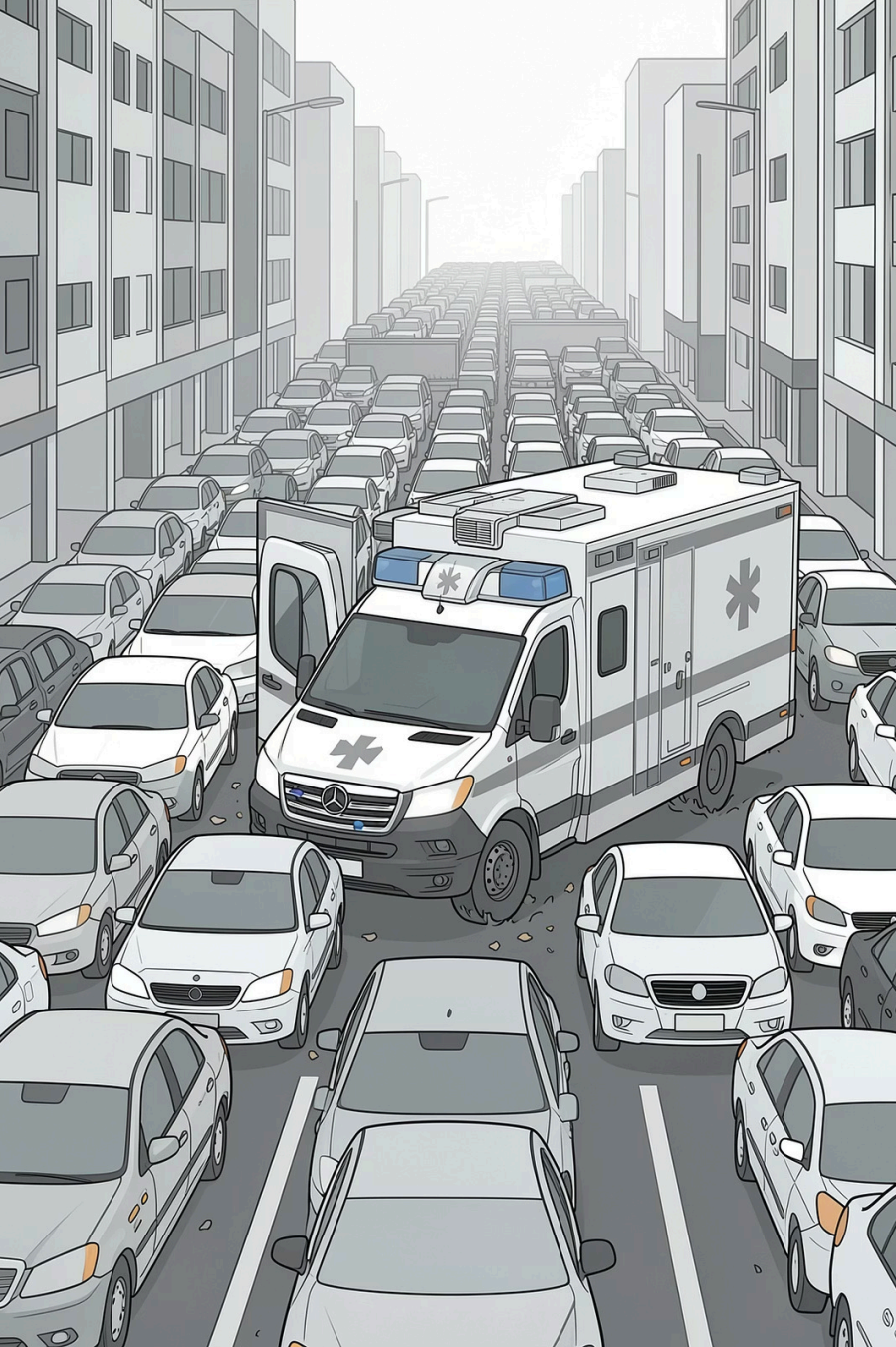


GreenWave

Helping ambulances move faster through Cairo traffic

Hamza Kamel · Khadeejah Iraky · Ziad Gaballah

CSCE 4301 Embedded Systems — Spring 2026



The Problem

Ambulances are stuck in traffic.

8 to 15 Minutes

Cairo ambulances take **8 to 15 minutes** to reach you.

10% Drop in Survival

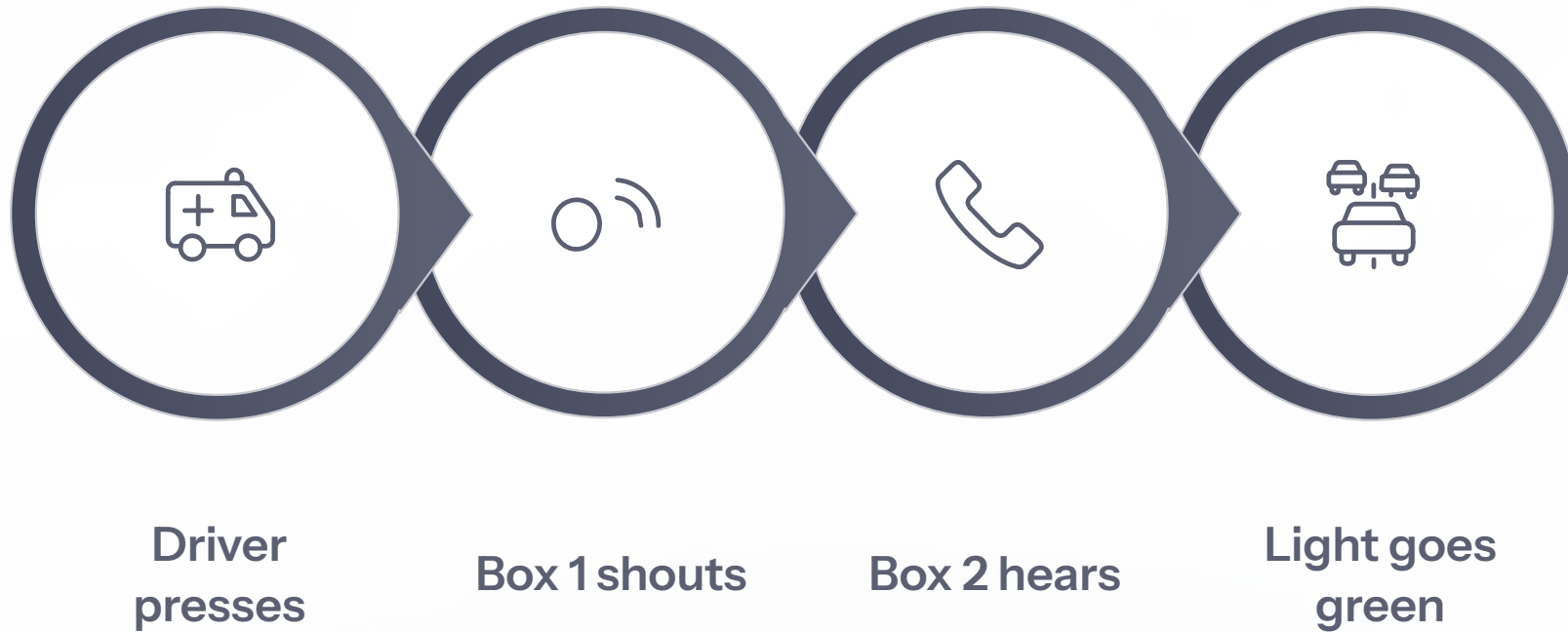
Every extra minute in cardiac arrest = **10% drop in survival**. That's why this isn't a "traffic" problem — it's a clinical one.

Zero Systems in Egypt

Egypt has **zero** systems that turn lights green for ambulances. The US, UK, Germany, and Dubai all have this. We don't.

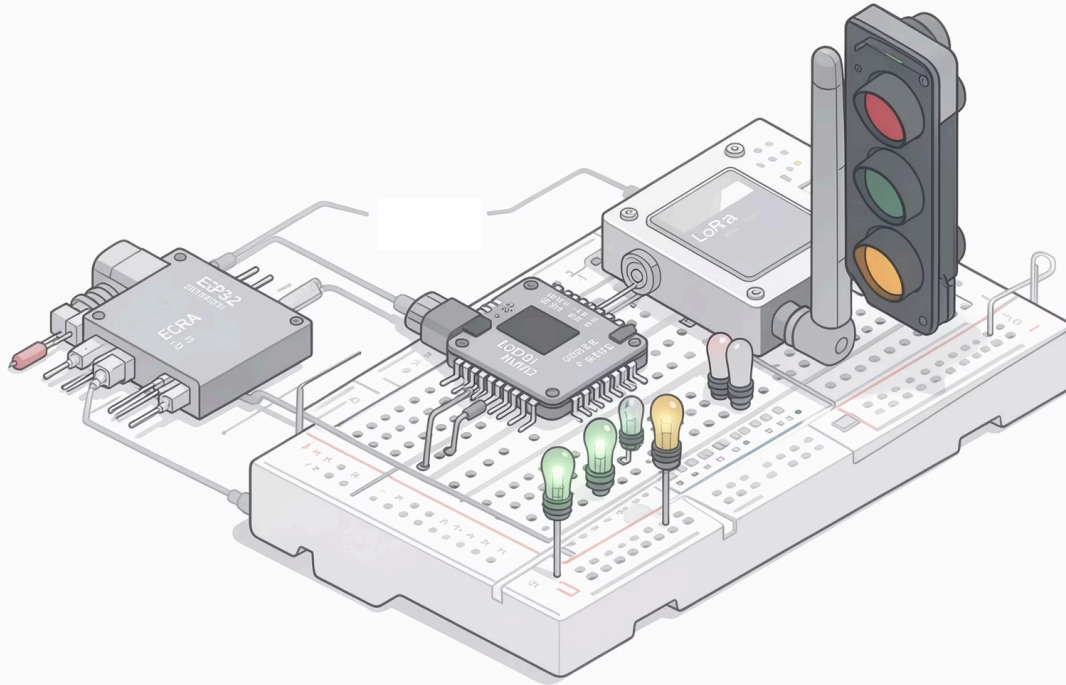
The Idea

Two boxes. One radio link. Lights turn green when an ambulance is coming.



Box 1 sits in the **ambulance**. Box 2 sits at the **traffic light**. Driver presses a button → box 1 shouts over radio → box 2 hears it → light goes green. Driver releases the button → light goes back to normal. That's it.

How It Looks — Block Diagram



System Architecture

The ambulance signs a message and sends it over LoRa radio. The intersection checks the signature, then turns its lights green. A second intersection is wired in over I²C so it follows along.

What's In Each Box

Both boxes use the same brain — an ESP32 — with different parts around it.

Ambulance Box

- ESP32
- Emergency button
- Status LED
- LoRa radio

Intersection Box

- ESP32
- Traffic LEDs
- LoRa radio
- I²C wire to next intersection

Why LoRa?

Long range (~1 km) and license-free at 433 MHz in Egypt.

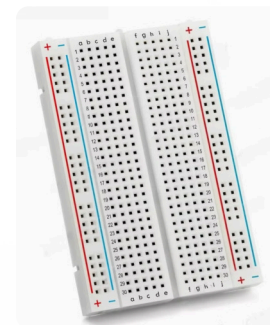
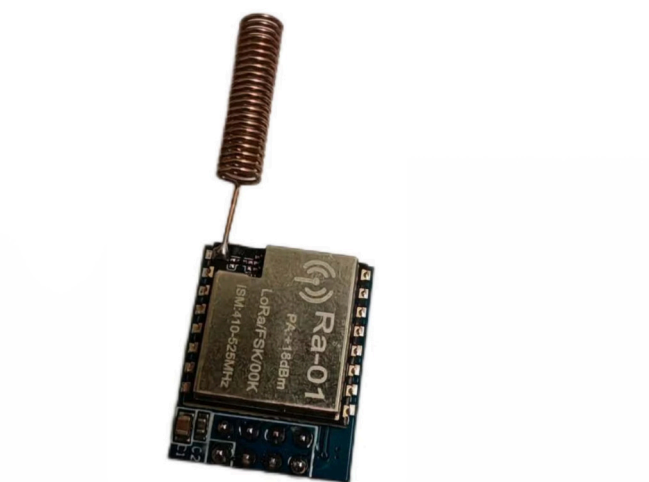
Why I²C?

Simple, reliable, classic embedded protocol — perfect for short wires between adjacent boxes.

Hardware Components

Everything we used, and what each part does:

- **ESP32 × 2** — the brain. Tiny microcontroller running FreeRTOS. One per box.
- **SX1276 LoRa Module × 2** — the long-range radio. Transmits/receives at 433 MHz, ~1 km range.
- **Traffic Light LED Module** — pre-built stack of red/yellow/green LEDs. One per intersection.
- **Push Button** — emergency button
- **Breadboards × 2** — one per box.
- **Jumper wires kit** — to connect everything.
- **Resistors** — 220 Ω for LEDs, 4.7 k Ω pull-ups for I²C bus.



How the Code is Organized

The ESP32 juggles four jobs at once using FreeRTOS — not one big loop.

Job	Rate	Why this rate
Watch the button	50 Hz	Fast enough to feel instant, slow enough not to waste CPU
Talk on the radio	2 Hz	One message every 500 ms — frequent enough to react, sparse enough not to clog the air
Run the lights	10 Hz	Smooth-looking transitions, no flicker
Stay safe (watchdog)	1 Hz	Catches a freeze within ~1 second

How They Talk

The ambulance shouts. The intersection listens. Nobody replies.

Ambulance → Intersection (LoRa 433 MHz, every 500 ms)

- Each message is **digitally signed** with HMAC-SHA256 → nobody can fake it
- Each message has a **sequence number** (a counter going up by 1) → nobody can record and replay it later

Why 433 MHz? It has a long wavelength (~69 cm) — the wave bends around buses, walls, and bodies. Higher-frequency radios bounce off everything.

Frequency vs. Range

Lower frequency = longer wave = bends around obstacles, travels further.

📄 That's why we called it **GreenWave** — the green spreads down the road.

What the Intersection Does

Six LEDs. Three for the main road, three for the side road.

1

Normally

Runs a regular traffic cycle — main green, then yellow, then side green, repeat.

2

Ambulance Arrives

When an ambulance message arrives: both sides flash yellow for 2 seconds, then **main goes green, side goes red** until the messages stop.

3

Ambulance Passes

3 seconds without a message: yellow transition again, back to normal cycle. (The ambulance sends every 500 ms, so 3 seconds = 6 missed packets in a row — enough to be sure it's actually gone, not just a brief radio dropout.)

4

Failure Mode

Anything breaks (power loss, bad signature, crash): all LEDs flash red — drivers treat it as a 4-way stop.

Knowing the Ambulance is Coming Closer

The radio signal gets stronger when the ambulance is approaching.

Signal Strength vs. Distance (RSSI)

Each message arrives with a number called **RSSI** (how strong the signal was, in dBm — closer to zero = stronger). Stronger over time → ambulance is **getting closer**. Weaker over time → ambulance is **leaving**.

